Sean Jackson

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Citizenship: US / UK / CA

Education

- **University of Toronto** — Toronto, ON
  Honours BSc. Physics; Minors in Computer Science & Mathematics — *September 2022 – May 2026*
  - Clubs: AI Club, Quantum Computing Club, Physics Student Union, Computer Science Student Union

Work Experience

- **Lawrence Livermore National Laboratory** — Livermore, CA
  MED Intern (Project APEX) — *May 2026 – Present*
  - **Agentic AI Workflows**: Build Python-based scientific AI agents for autonomous materials laboratories, defining structured tools, prompt logic, validated outputs, and multi-step task workflows over documented equipment and platform interfaces.
  - **Tool & Equipment Integration**: Integrate AI agents with scientific software, internal APIs, databases, cloud services, and PLC-controlled equipment to execute experiments and coordinate closed-loop workflows from planning through data collection and analysis.
  - **Evaluation & Reliability**: Evaluate and debug multi-component AI workflows using structured logging, traceable tool calls, repeatable test cases, and failure analysis to improve prompt behavior, tool selection, and end-to-end reliability.
  - **Safe Physical Automation**: Collaborate with controls, platform, and deployment engineers to translate operational tasks into maintainable AI-assisted workflows with constrained action boundaries, approval points, error handling, and fallback behavior.
  - **Software & Platform Engineering**: Develop Python and Golang services, APIs, and message-driven integrations connecting foundation models, scientific applications, telemetry systems, cloud services, and laboratory automation infrastructure.
  - **Deployment & Reproducibility**: Deploy containerized AI/ML applications using AWS and Docker, emphasizing configuration management, reproducible execution, monitoring, and maintainable software interfaces.
  - **Model Development & Deployment**: Build, evaluate, and deploy PyTorch-based machine-learning systems for scientific applications, spanning dataset engineering, model development, inference, and integration into automated experimental workflows.

- **Lawrence Livermore National Laboratory** — Livermore, CA
  Machine Learning Engineer (Data Science Scholar Institute, Summer Intern) — *May 2025 – September 2025*
  - **Project APEX**: Designed, trained, and deployed deep computer-vision models for high-throughput materials characterization in a self-driving laboratory for automated alloy discovery; curated training datasets in collaboration with Cornell University.
  - **ML Deployment**: Developed end-to-end PyTorch computer-vision pipelines spanning real-world image curation, model architecture, training, quantitative evaluation, and deployment in limited-compute experimental environments.
  - **Robotics & Automation**: Built low-level robotics and tooling frameworks integrated into higher-level architectures for closed-loop laboratory automation (*details restricted under U.S. DOE regulations*).
  - **CV Design Challenge**: Designed generative decoding models using temporal feature representations for tracking occluded objects through walls.

Publications

- **First Author: Classifying Alloy Surface Preparation Quality with Metadata Infused Machine Learning for Rapid Alloy Discovery**: Under review. Developed a novel end-to-end computer-vision framework for automated materials assessment in self-driving laboratories. Led model design, implementation, and evaluation using imaging data from Cornell University with validation at LLNL.

Projects – GitHub

- **GRACE: GRinding Automated Classification Engine** GitHub
  Custom PyTorch computer-vision framework for automated materials quality assessment in self-driving laboratories. Associated first-author research paper currently under review.

- **SmartQueue** GitHub | Website
  UofT office-hours queue system with dynamic wait-time estimation, semantic request clustering using DBSCAN and SBERT, and hybrid in-person/remote support.

- **AT-AT-Inspired Reinforcement Learning Locomotion**
  Built an articulated quadruped physics simulation in MuJoCo and trained a reinforcement-learning policy for straight-line locomotion by designing the robot dynamics, observations, actions, reward function, and evaluation workflow.

Skills

- **AI & Multimodal Systems**: PyTorch, computer vision, multimodal perception, LLMs, vision-language models, Transformers, reinforcement learning, dataset engineering, model training, quantitative evaluation, model deployment, Triton, Megatron-LM.

- **AI Agents & Evaluation**: Multi-step agent workflows, tool/function calling, prompt logic, structured outputs, retrieval-augmented generation (RAG), workflow evaluation, guardrails, failure analysis, human-in-the-loop approval, autonomous experimentation.

- **Backend & Software Engineering**: Python, Golang, C/C++, Java, SQL, Git, REST/API development, message-driven systems, software architecture, distributed systems, CI/CD, GitHub Actions, Jenkins.

- **Cloud & Infrastructure**: AWS, Docker, containerized applications, cloud services, distributed computing, edge and limited-compute deployment.

- **Physical AI & Laboratory Automation**: PLC integration, automated laboratory equipment, telemetry data, closed-loop control, hardware/software interfaces, MuJoCo simulation, ROS 2 exposure, hardware-in-the-loop workflows, Arduino, Raspberry Pi.

- **High-Performance Computing**: GPU computing, distributed ML workloads, large-scale simulation, NVIDIA and AMD supercomputing systems, Sierra, Lassen.

- **Scientific Computing**: MATLAB, scientific data processing, experimental automation, autonomous scientific workflows.

- **Quantum Computing**: Qiskit, Cirq, quantum algorithms, quantum information theory, quantum key encryption.